

To: Dr. Pengcheng Xi Pengcheng.xi@nrc-cnrc.gc.ca

Week 3 Report

Yiran Liu

NRC - Digital Technology, 27.01.2026

1. Work Completed Since Last Meeting

1.1. Pipeline Integrity Check (Continued)

In the meeting last week, I was advised to replace the Step2 -> Step3 code with the official implementation from the HumanML3D research.

I conducted an analysis of the generated HumanML3D skeleton data by rendering a shape-normalized SMPL mesh model. I found that, in addition to potential defect due to the “inflated shape problem”, the dataset has **more severe problems**:

- Floating Artifacts: Subjects appear meters above the ground plane
- Inverted Motion: Subjects moving backwards due to incorrect root rotation

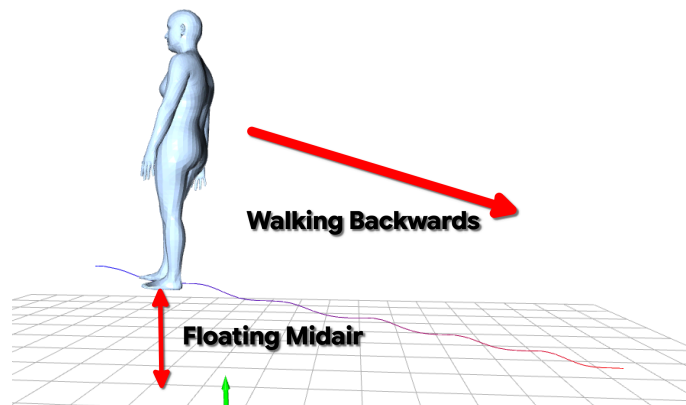


Figure 1: Critical artifacts in Step 3 data (floating & inverted motion)

The conclusion is clear: the current Step 2 $\rightarrow$ 3 conversion code from the previous student is critically flawed, **and cannot be used for production**.

1.2. MoSh++ and HumanML3D Codebase Containerization

To replace the broken pipeline, I proposed to use MoSh++ - the gold standard for AMASS, and scripts from the HumanML3D research.

However, both codebases are outdated by 6-7 years, this has posted significant challenges:

- **Dependency Conflicts:** Python 2-era libraries incompatible with modern environments
- **Build Failures:** End-Of-Life (EOF) software versions require direct compilation from C++ source code.

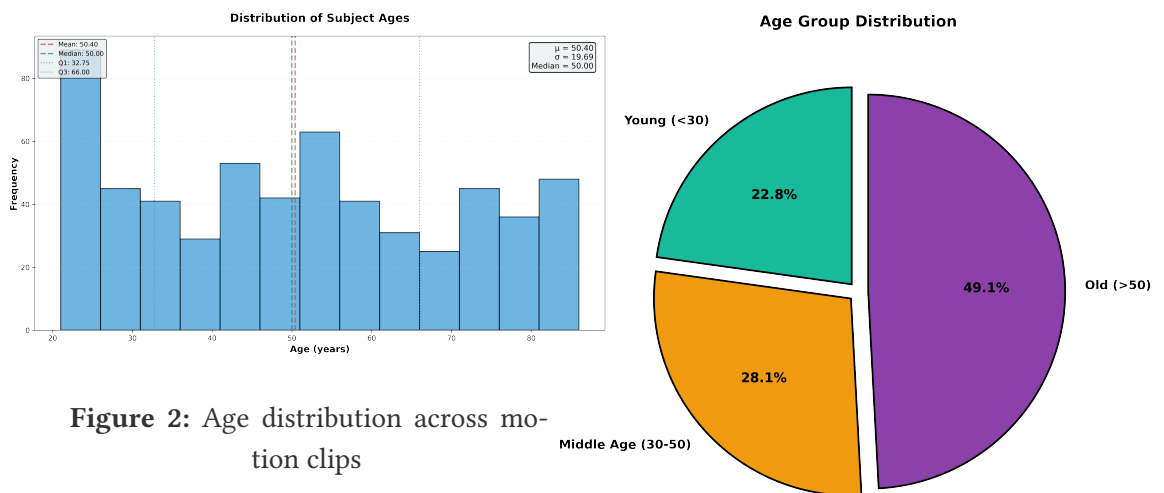
Solution: The Docker container is now complete and validated

After extensive engineering effort, I compiled dependencies from source and built a sharable Docker Image based on Ubuntu20.04. This solution permanently resolves dependency issues and ensures reproducibility for future researchers.

However, this has delayed the project timeline by at least two days.

1.3. Dataset Splitting: Age Categorization

In the mean time, I parsed metadata from the Van Crielinge dataset and implemented splitting scripts to categorize subjects into age groups. The distribution analysis is shown below:



A python script was written split dataset into balanced age groups, it will be in application once motion data is re-processed.

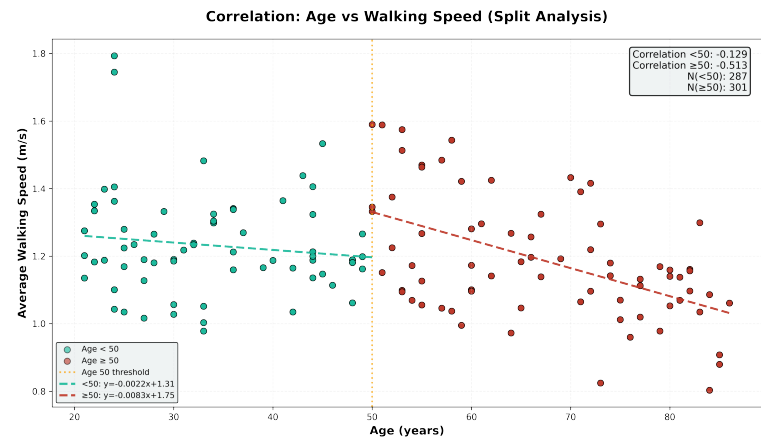


Figure 4: Walking speed correlation with age. A strong regression correlation is observed for subjects aged 50+, confirming the previous observation that walking speed decreases with age.

1.4. Documentation

I’ve compiled an 8-page Technical Analysis detailing pipeline failures and the mathematics behind coordinate system errors. This document is available in the repository for reference.

2. Project Timeline & Adjustments

Week 1	Onboarding, foundational study, project structure (Completed)
Week 2	Dataset analysis, pipeline investigation (Completed)
Week 3	Pipeline audit, MoSh++ containerization (Completed)
Week 4	Age-based data splitting (Completed), Pipeline diagnosis (Completed), STGCN baseline training (Revised)
Week 5	Train three distinct LoRA adapters: Young, Adult, and Elderly
Feb 15	Submit extended abstract (Target)
Week 6-7	Fine-tuning, validation experiments, metrics
Week 8-9	Full paper writing, supplemental experiments

Red indicates postponed tasks; Purple indicates revised tasks

Timeline remains tight but achievable. I will continue to aim at our Feb 15th abstract target.